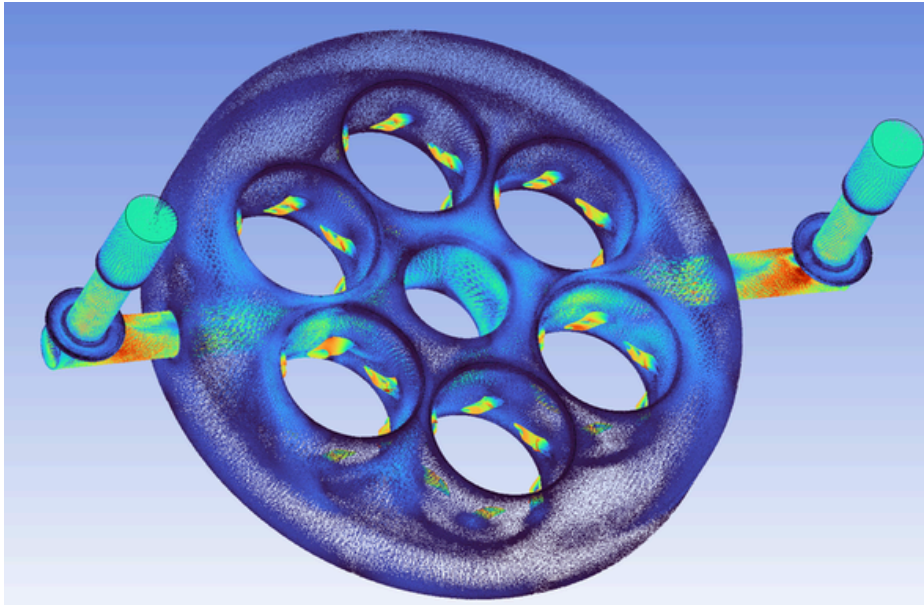


- High-fidelity CFD Simulation meshed in ANSYS Fluent with a target y^+ of 1.0.
- Utilized ANSYS Fluent's steady-state RANS $k-\omega$ model.

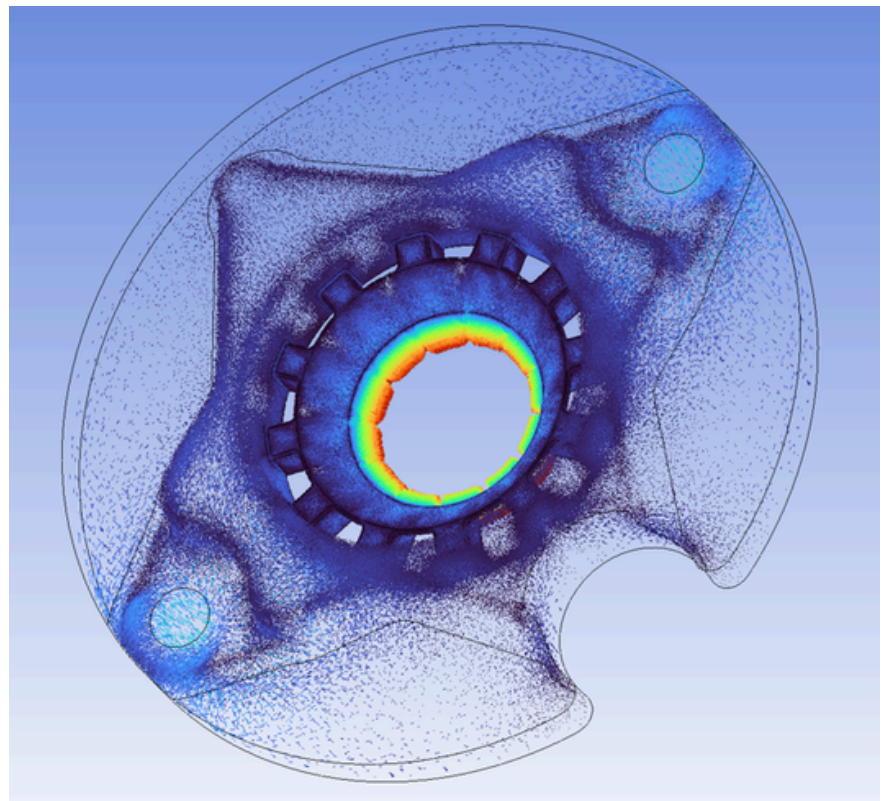


Coaxial Swirl Injector Manifold

- Analyzed flow maldistribution and pressure concentrations across the manifold elements.
- Analyzed flow concentrations through our manifold inlets, revealing critical velocity concentrations along the bends.
- Reconfigured the manifold elements, varying swirl direction and tangential inlet position to alleviate pressure concentrations.

Pintle Fuel Injector Manifold

- Analyzed flow maldistribution and pressure concentrations across slots feeding into a pintle injector annulus.
- Validated structural slots as pintle locating/supporting features and confirmed that internal pocketing weight reduction had negligible effect on discharge coefficient.
- Estimated a discharge coefficient of 0.60, aligning with the discharge coefficient tested in last year's pintle design.



Results

- An estimated 7% more maldistribution in the coaxial swirl injector manifold than in the pintle injector manifold and excessive velocity in our inlets prompted the team to select the pintle injector design.

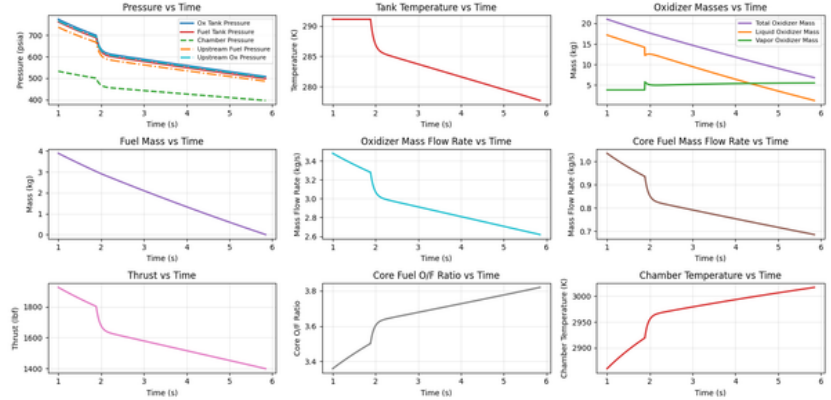
VERTIGO TWO-STAGE SIMULATION -



SPRING 2026

- Programmed a supercharged (back-pressured) to saturated blowdown simulation for an Ethanol/N₂O engine in Python.
- Modeled isentropic ullage expansion with GN₂ back-pressurant through the supercharged phase.
- Modeled nitrous vaporization for self-pressurization through the saturated phase.
- Optimized ullage volume, back-pressure, and propellant mass to maximize total impulse and supercharged time for fixed state points.

Two-stage Blowdown Results



Results

- Expansion through the supercharged phase catastrophically decreased saturated tank pressure and temperature, while also requiring massive run-tanks for sufficient ullage, prompting the team to select an exclusively saturated engine.

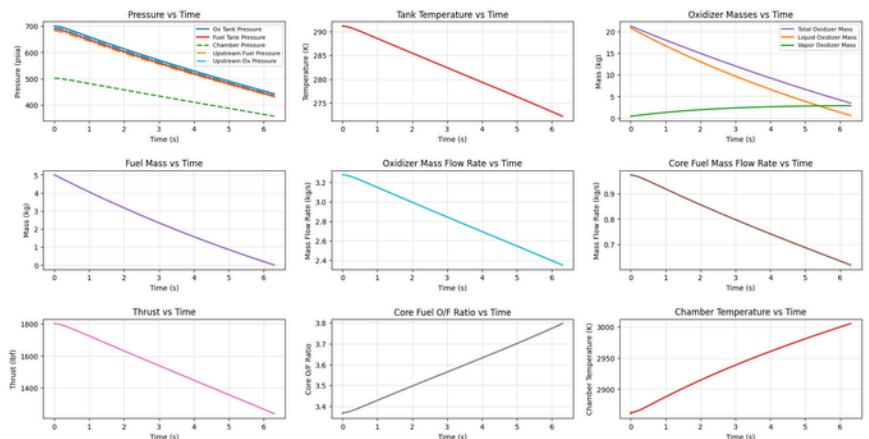
VERTIGO ENGINE SIMULATION RE -



SPRING 2026

- Owned an entirely saturated Ethanol/N₂O engine blowdown simulation in Python.
- Calculated total impulse to optimize nitrous oxide state points and propellant masses for an 8 TWR target.
- Implemented a module to size Ethanol and nitrous inlet holes, incorporating raceway pressure loss and the Dyer two-phase flow model.

Nitrous Oxide Blowdown Results



Results

- Utilized engine simulation to design thrust chamber characteristics, optimizing for an apogee target of 20,000 ft, a burn-time limitation of 7 s, and a chamber temperature ceiling of 3050 K.
- Optimized nozzle parameters based on projected apogee coupled to our thrust curves.

